Cohort, labels, preprocessing, architecture and optimizer are identical across all three arms

Cohort
740 patients
1,695 nodules
11,026 slices

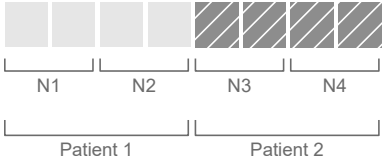
Preprocessing
50% consensus ROI
256 × 256 px
identical in all arms

Backbone
ImageNet-pretrained
DenseNet-121 · EfficientNet-B0 · ViT-S/16

only the partition unit is manipulated

(a) Arm A — Patient-grouped

GroupKFold on patient



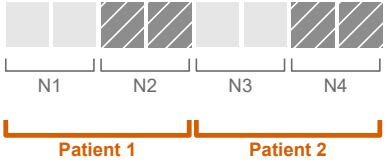
Fraction of test rows that share a

patient with the training fold	0.000
nodule with the training fold	0.000

neither unit crosses the boundary

(b) Arm C — Nodule-grouped

GroupKFold on nodule



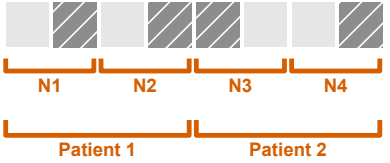
Fraction of test rows that share a

patient with the training fold	**0.690**
nodule with the training fold	0.000

patient route on, within-nodule route off

(c) Arm B — Random

KFold, no grouping



Fraction of test rows that share a

patient with the training fold	**0.995**
nodule with the training fold	**0.973**

both routes on

B − A total gap
C − A patient route
B − C within-nodule route
training slice
test slice
unit split across training and test